

Southeast University Examination Paper (A)

Course Name Principles of Compiling Examination Term 11-12-2 Score _____
Related Major Computer Science & Technology Examination Form Close test Test Duration 150 Mins

There are 8 problems in this paper. You can write the answers in English or Chinese on the attached paper sheets.

1. Please construct context-free grammars **with ϵ -free productions** for the following language (10%).

$\{\omega \mid \omega \in (a,b,c,d)^* \text{ and the numbers of } a\text{'s and } b\text{'s and } c\text{'s occurred in } \omega \text{ are odd, and } \omega \text{ starts with } a, \text{ ends with } b \text{ or } c\}$

2. Please construct a **DFA with minimum states** for the following regular expression. (15%)

$((a|b)^*(ab)^*)^*(a|b)$

3. Please **eliminate the left recursions (if there are)** and **extract maximum common left factors (if there are)** from the following context free grammar, and then decide **the resulted grammar** is whether a LL(1) grammar by **constructing the related LL(1) parsing table.**(15%)

$S \rightarrow \text{begin } L \text{ end} \mid \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid \text{while } E \text{ do } S \mid a$

$L \rightarrow L; S | S$

$E \rightarrow E \text{ and } E | E \text{ or } E | b$

4. Please **construct a LR(1) parsing table for the following ambiguous grammar with your own defined additional conditions (You determine the required additional conditions by yourself).**(15%)

$S \rightarrow \text{if } E \text{ then } S | \text{if } E \text{ then } S \text{ else } S | a$

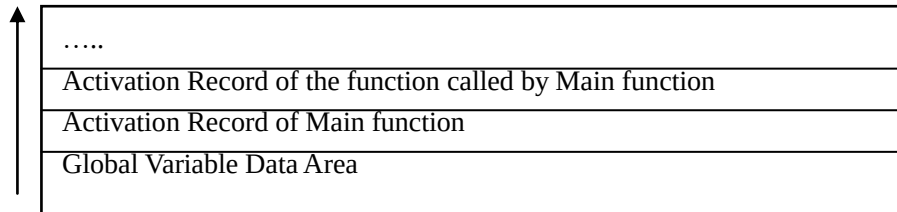
$E \rightarrow E \text{ and } E | E \text{ or } E | b$

5. Please construct **an annotated parse tree** for the input string 101.101 where the syntax-directed definition is as following (10%):

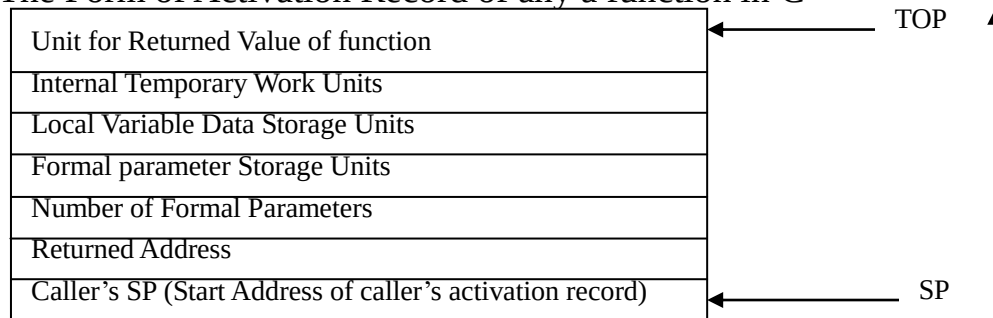
Productions	Semantic Rules
$S \rightarrow L_1.L_2$	$\{S.val = L_1.val + L_2.val / 2^{L_2.len}\}$
$S \rightarrow L$	$\{S.val = L.val\}$
$L \rightarrow L_1B$	$\{L.val = L_1.val * 2 + B.val, L.len = L_1.len + 1\}$
$L \rightarrow B$	$\{L.val = B.val, L.len = 1\}$
$B \rightarrow 0$	$\{B.val = 0\}$
$B \rightarrow 1$	$\{B.val = 1\}$

6. We assume that the storage organization and the form of activation record used in C language program run-time stack storage allocation are as following. Please **construct the run-time stack map when it gets the maximum size** for the following C program (10%).

Storage Organization of C Language



The Form of Activation Record of any a function in C



```
#include <stdio.h>
```

```
int x,y,z;
```

```
int main()
```

```
{
    x=10;
    y=6;
    z=C(x,y);
}
```

```
int C(int m, int n)
```

```
{
    if (m>=n)
        if (n<=0) return 1;
        else if (n==1) return m;
```

```

else {
    int t1;
    t1=m-n;
    if (t1<n)
    {
        int t2;
        t2=C(m,t1);
        return t2
    }
    else {
        int t3,t4,t5,t6,t7;
        t3=m-1;
        t4=n-1;
        t5=C(t3,t4);
        t6=m*t5;
        t7=t6/n;
        return t7;
    }
}
else return 0;
}

```

Notes: 1) Here we assume that the caller's sp of Main function is the start address of global variable data area, and the returned address in the activation record of a function (including Main function) is filled by the operating system automatically, you might not care it.

2) The initial value of variable X is 10 and the initial value of variable Y is 6, the start address of stack used in the program is K.

7. Please translate the following program fragment into **three-address-code (TAC) sequence using short circuit code and back-patching techniques**. (15%)

```

i=1;
loop=0;
while (loop==0 && i<=10) {
    j=1;
    while (loop ==0 && j<=20)

```

```

        if (a[i,j] != b[i,j])
        {
            loop=1;
            m=i;
            n=j;
        }
        else j=j+1;
        if (loop==0) i=i+1;
    };
    if (loop==1) e=0;
    else e=1;

```

Notes: Here we assume that the declaration of array A and B are array [1..10,1..20], each data element of array A or B would **use 4 storage unit**, and the start address of array A's storage area is addrA, the start address of array B's storage area is addrB.

8. Please **construct the DAG** for the following basic block, optimize the block and **rewrite the block** in optimized code form. Note that we assume **only Variable L would be used later**(10%)

```

B=3
D=A+C
E=A*C
F=D+E
G=B*F
H=A+C
I=A*C
J=H+I
K=B*5
L=K+J
M=L

```